

★ WEEK 9 · PART 1 · PRACTICE

Predict and Update, Worked by Hand

Follow One Surprise Through the Filter; Read the Innovation

The predict-update cycle, made concrete on a tracked price

Part 1's mechanics, with the arithmetic shown. We walk one full predict-update iteration by hand - predicted observation, innovation, gain, correction - then use the standardized innovation to flag news and broken models.

1P

PART 1 · APPLICATIONS

The Surprise That Runs the Filter

Case A: one full predict-update iteration on a tracked fair value, with the innovation and its variance made explicit.
Case B: the standardized innovation as a signal - flagging a four-sigma surprise. Then a Claude Code lab.

What Part 1 Told Us

Now we run one step - with numbers

1

Predict, then update

Move the state and its uncertainty forward, then correct them when the new observation arrives. Two steps, repeated forever.

2

The innovation is the surprise

The gap between what you observed and what you predicted. It drives every correction - no surprise, no update.

3

The gain weighs the surprise

How much to trust the surprise depends on the prediction and measurement uncertainty. We compute one full step by hand.

Everything here is a calculator exercise: one prior, one print, one surprise - then four of them.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a prior state and its variance.
- Compute predict, innovation, and update by hand.
- Then read what the surprise is telling you.

What you should watch

- How the gain splits the difference to the print.
- How the innovation variance sets the gain.
- How a big standardized innovation raises a flag.

One Predict-Update, Start to Finish

Follow a single surprise through the filter

A desk tracks a fair value that drifts. A new print arrives. We walk the full cycle by hand: predict the state and its variance forward, predict the observation, measure the innovation and its variance, form the gain, and correct the estimate. Every piece of the Kalman filter in one step

The ingredients

- Prior state 100.0, variance 0.20.
- Process (drift) variance $Q = 0.05$.
- Measurement noise $R = 0.30$.

What we'll do by hand

- Predict the state and grow the variance.
- Compute the innovation and its variance.
- Form the gain and update the state.

The Predict Step

Move the state and uncertainty forward

predict state: $x^- = x = 100.0$

predict variance: $P^- = P + Q = 0.20 + 0.05 = 0.25$

With no new data yet, the state stays at 100.0 but its variance grows by the drift Q - the filter gets less certain as time passes.

The Innovation and Its Variance

The surprise, and how surprising it is

$$\begin{aligned} \text{predicted obs: } \hat{y} = x^- &= 100.0, & \text{actual } y &= 100.8 \\ \text{innovation: } \nu = y - \hat{y} &= +0.8 & & \text{(the surprise)} \\ \text{innovation var: } S = P^- + R &= 0.25 + 0.30 = 0.55 \end{aligned}$$

The print beat the prediction by 0.8 - that's the innovation. Its variance $S = 0.55$ combines how unsure we were with how noisy the print is.

The Update Step

Correct the state by the gain

$$\text{gain: } K = \frac{P^-}{S} = \frac{0.25}{0.55} = 0.455$$

$$\text{update state: } x = x^- + K \nu = 100.0 + 0.455(0.8) = \mathbf{100.36}$$

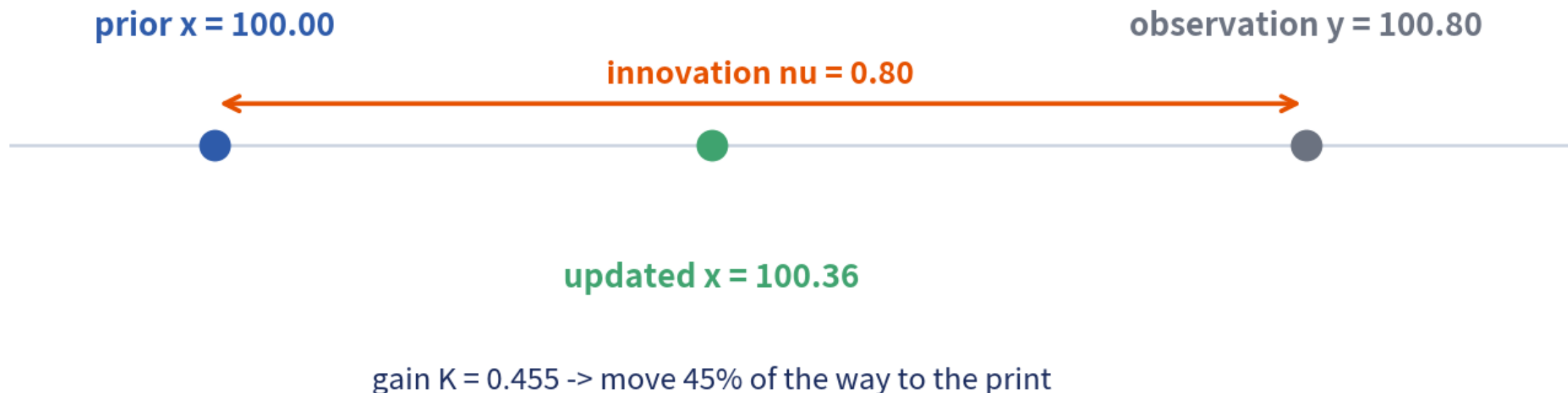
$$\text{update variance: } P = (1 - K) P^- = 0.14$$

The gain 0.455 moves the estimate 45% of the way to the print - to 100.36 - and the variance shrinks to 0.14. The surprise has been absorbed.

One Iteration, on a Line

The prior sits at 100.00 and the print at 100.80 - an innovation of 0.80. The gain of 0.455 lands the updated estimate at 100.36, about 45% of the way to the print. A bigger gain would trust the print more; a smaller one, less.

Case A: one predict-update - the innovation pulls the estimate part-way to the print



The Innovation Is the News

Every update is driven by surprise

No surprise, no update

- If the print equals the prediction, the innovation is zero.
- The gain has nothing to act on.
- The estimate doesn't move.

The surprise carries the information

- Only the unexpected part of a print is new.
- The gain decides how much of it to believe.
- That is the whole logic of the filter.

The Innovation as a Signal

When a surprise is too big to ignore

The innovation isn't just fuel for the update - it's information in its own right. Standardize it by its own standard deviation, and you get a z-score for surprise. A normal-sized one is business as usual; a four-sigma one means real news, or a model that has broken. We track four steps by hand

The idea

- Standardize the innovation: $z = nu / \text{sqrt}(S)$.
- It measures surprise in standard deviations.
- Most steps sit comfortably within +/- 2.

What we'll do by hand

- Run four predict-update steps.
- Compute the standardized innovation each time.
- Flag the one that breaks the +/- 2 band.

The Standardized Innovation

Surprise, measured in sigmas

$$z = \frac{\nu}{\sqrt{S}} = \frac{0.8}{\sqrt{0.55}} = 1.08 \Rightarrow \text{a normal-sized surprise } (|z| < 2)$$

How to read it

- z near zero: the print matched the model.
- $|z|$ up to 2: normal, expected noise.
- $|z|$ above 2: a genuinely large surprise.

Why standardize

- Raw innovations aren't comparable over time.
- S changes as the filter's certainty changes.
- Dividing by $\text{sqrt}(S)$ makes them a clean z -score.

Four Steps of Surprise

Standardized innovation each step

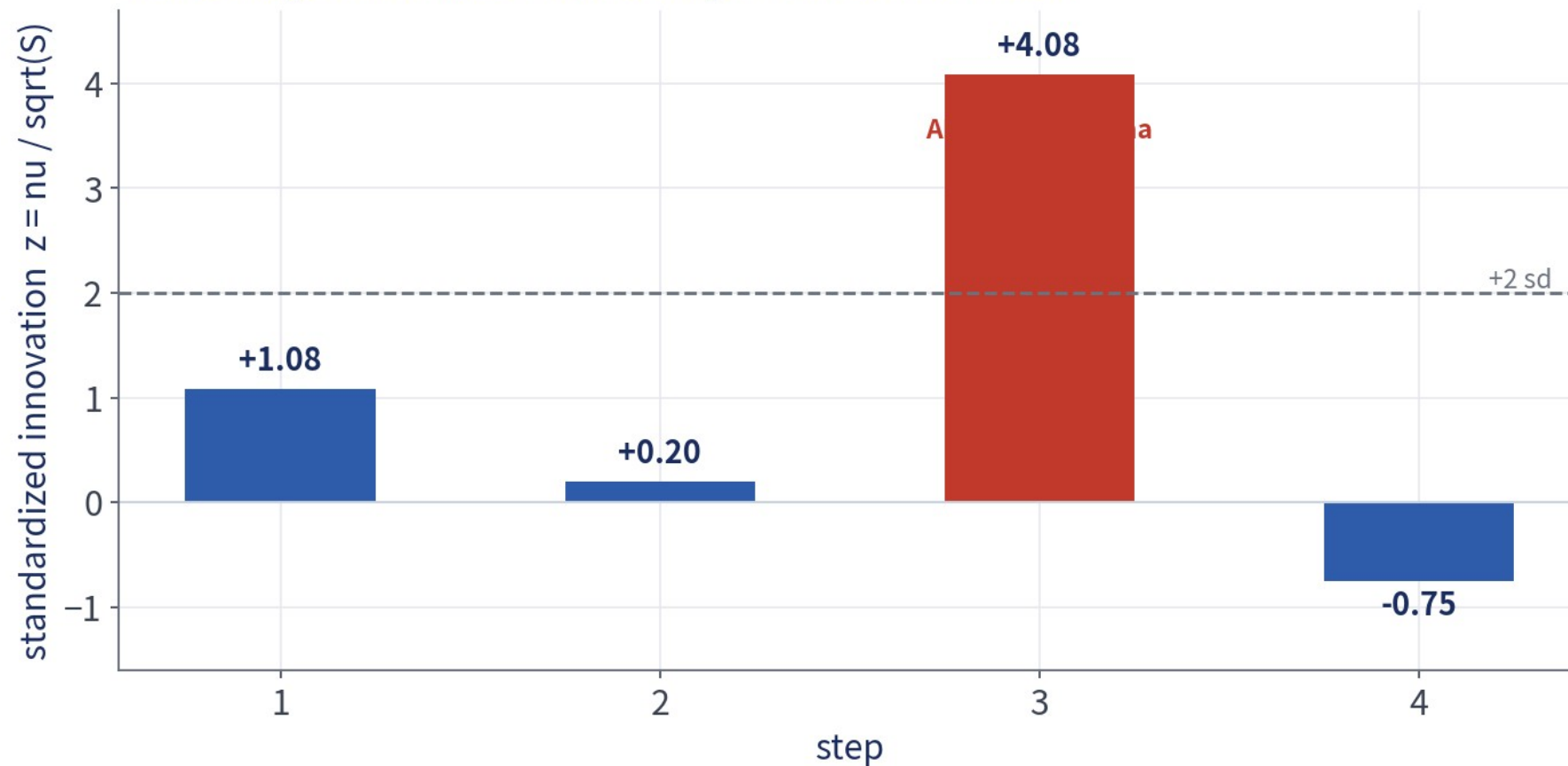
step	print y	innovation nu	S	$z = \text{nu} / \text{sqrt}(S)$
1	100.8	+0.80	0.55	+1.08
2	100.5	+0.14	0.49	+0.20
3	103.2	+2.78	0.47	+4.08 ALARM
4	100.9	-0.50	0.46	-0.75

Steps 1, 2, and 4 are ordinary. Step 3's print of 103.2 is a four-sigma surprise - far beyond what the filter expected.

A Four-Sigma Surprise

Three of the four standardized innovations sit inside the ± 2 band - normal noise. Step 3 spikes to +4.08, far outside it. That is the filter telling you something real happened: a news event, a data error, or a state that has genuinely jumped

Case B: a large standardized innovation flags news or a broken model



What a Big Surprise Means

News, error, or a broken model

As an alarm

- A large $|z|$ flags a structural break or news.
- Or a bad data point that needs checking.
- The filter's own math raised the flag.

As a signal

- On a mean-reverting spread, a big z can be noise.
- If so, fade it - trade against the surprise.
- Either way, the innovation is telling you to act.

The Common Pattern

The surprise runs everything

Both cases turn on the innovation. In the update, it's the fuel: the estimate moves only because the print surprised us, scaled by the gain. As a signal, its standardized size tells us whether the surprise is ordinary noise or something that demands attention. Predict, be surprised, respond

The recipe

- Predict the observation and its variance.
- Measure the innovation - the surprise.
- Update by the gain, or flag if too large.

Why it's powerful

- The same number drives learning and alerting.
- The gain adapts to how noisy things are.
- Nothing moves without a surprise.

Reproduce the Iteration, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify the predict-update step and the innovation monitor, then run both on real data.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - one iteration:

- * from $x=100.0$, $P=0.20$, $Q=0.05$, $R=0.30$, $y=100.8$,
print the predict, innovation, S , gain, and update;

CASE B - innovation monitor:

- * run four steps on $[100.8, 100.5, 103.2, 100.9]$;
- * print nu , S , and $z = nu/\sqrt{S}$ each step;
- * flag any $|z| > 2$.

Then run the filter on a real price series and plot the standardized innovations with ± 2 bands.

Match every printed number to the slides.

Check against the slides

- predict $P_- = 0.25$.
- innovation 0.80 , $S = 0.55$.
- gain 0.455 , update 100.36 .
- step 3: $z = +4.08$ (alarm).

Deliverable: `part1_practice.py` whose printed numbers match the worked iteration and the z run.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What is the innovation?

**Observed minus
predicted - the
surprise**

Q2 What does the gain do to the surprise?

**Scales how much of
it corrects the state**

Q3 What does a large standardized innovation mean?

**News, a data error, or
a broken model**

All three come from the predict-update iteration and the standardized-innovation monitor.

Three Pitfalls to Avoid

Where predict-update slips

Common mistakes

- "Forget to grow the variance in predict."
- "Compare raw innovations across time."
- "Treat every big surprise as a trade."

The reality

- Predict adds Q - the filter loses certainty over time.
- Standardize by $\sqrt{S}$ before comparing surprises.
- A big z may be news or error - check before acting.

Two Calculations to Remember

Part 1 applications in one screen

1

One predict-update, by hand

Predict grew the variance to 0.25; the print's innovation was 0.80 with variance 0.55; the gain 0.455 moved the estimate to 100.36 and shrank the variance to 0.14. The surprise, scaled by the gain, is the whole update.

2

The standardized innovation is a signal

Dividing the innovation by $\sqrt{S}$ gives a z-score for surprise. Three steps sat inside ± 2 ; step 3's print of 103.2 spiked to $z = +4.08$ - a flag for news, error, or a broken model.

3

Nothing moves without a surprise

The innovation drives the update and doubles as an alarm. Predict, measure the surprise, and respond - by correcting the state, or by raising a flag.

Next — Part 2 practice: the Kalman gain itself, and how it balances trust between model and data.